APPENDIX C—CURRENTLY ASSIGNED SUBSTANCES

C.1 Self-Reactive Substances of Division 4.1

This list is based on paragraph 2.4.2.3.2.4 of the UN Model Regulations, with irrelevant material removed.

Notes:

1. *Self-reactive substances to be transported must fulfil the classification and the control and emergency temperatures (derived from the SADT) as listed.*
2. *Self-reactive substances not listed in Table C.1 are subject to classification approval by the appropriate national authority of the State in which the dangerous goods were manufactured (See 3.4.1.2.4.1).*

TABLE C.1
List of Currently Assigned Self-Reactive Substances of Division 4.1 in Packages

Self-Reactive Substance	Concentration (%)	Control Temperature (°C)	Emergency Temperature (°C)	UN Generic Entry	Notes
Acetone-pyrogallol copolymer 2-diazo-1-naphthol-5-sulphonate	100			3228	
Azodicarbonamide, formulation type B, temperature controlled	< 100			Forbidden	1, 2
Azodicarbonamide, formulation type C	< 100			3224	1
Azodicarbonamide, formulation type C, temperature controlled	< 100			3234	1
Azodicarbonamide, formulation type D	< 100			3226	1
Azodicarbonamide, formulation type D, temperature controlled	< 100			3236	1
2,2'-Azodi (2,4-Dimethyl-4-methoxyvaleronitrile)	100	-5	+5	3236	
2,2'-Azodi (2,4-Dimethyl-valeronitrile)	100	+10	+15	3236	
2,2'-Azodi (Ethyl 2-methylpropionate)	100	+20	+25	3235	
1,1'-Azodi (Hexahydrobenzonitrile)	100			3226	
2,2'-Azodi (Isobutyronitrile)	100	+40	+45	3234	
2,2'-Azodi (Isobutyronitrile), as a water based paste	≤ 50			3224	
2,2'-Azodi (2-Methylbutyronitrile)	100	+35	+40	3236	
Benzene-1,3-Disulphonylhydrazide, as a paste	52			3226	
Benzenesulphonyl hydrazide	100			3226	
4-(Benzyl(methyl)amino)-3-ethoxybenzenediazonium zinc chloride	100	+40	+45	3236	
4-(Benzyl(ethyl)amino)-3-ethoxybenzenediazonium zinc chloride	100			3226	
3-Chloro-4-Diethylaminobenzenediazonium zinc chloride	100			3226	
2-Diazo-1-Naphthol-4-Sulphonyl chloride	100			Forbidden	2
2-Diazo-1-Naphthol-5-Sulphonyl chloride	100			Forbidden	2
2-Diazo-1-Naphthol sulphonic acid ester mixture, type D	< 100			3226	5
2,5-Dibutoxy-4-(4-Morpholinyl)-Benzenediazonium, tetrachlorozincate (2:1)	100			3228	
2,5-Diethoxy-4-Morpholinobenzenediazonium tetrafluoroborate	100	+30	+35	3236	
2,5-Diethoxy-4-Morpholinobenzenediazonium zinc chloride	67–100	+35	+40	3236	
2,5-Diethoxy-4-Morpholinobenzenediazonium zinc chloride	66	+40	+45	3236	
2,5-Diethoxy-4-(4-Morpholinyl)-Benzenediazonium sulphate	100			3226	
2,5-Diethoxy-4-(Phenylsulphonyl) Benzenediazonium zinc chloride	67	+40	+45	3236	
Diethylenglycol bis (allyl carbonate) + Di-isopropyl-peroxydicarbonate	≥ 88 + ≤ 12	-10	0	3237	
2,5-Dimethoxy-4-(4-Methylphenylsulphonyl) Benzenediazonium zinc chloride	79	+40	+45	3236	
4-(Dimethylamino)-Benzenediazonium trichlorozincate (-1)	100			3228	
4-Dimethylamino-6-(2-Dimethylaminoethoxy) toluene-2-Diazonium zinc chloride	100	+40	+45	3236	
N,N'-Dinitroso-N,N'-Dimethyl terephthalamide, as a paste	72			3224	
N,N'-Dinitrosopentamethylenetetramine	82			3224	3
Diphenyloxide-4,4'-Di-sulphonyl hydrazide	100			3226	

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C.1

TABLE C.1
List of Currently Assigned Self-Reactive Substances of Division 4.1 in Packages *(continued)*

Self-Reactive Substance	Concentration (%)	Control Temperature (°C)	Emergency Temperature (°C)	UN Generic Entry	Notes
4-Dipropylaminobenzenediazonium zinc chloride	100			3226	
2-(N,N-Ethoxycarbonylphenylamino)-3-Methoxy-4-(N-Methyl-N-Cyclohexylamino) Benzenediazonium zinc chloride	63–92	+40	+45	3236	
2-(N,N-Ethoxycarbonylphenylamino)-3-Methoxy-4-(N-Methyl-N-Cyclohexylamino) Benzenediazonium zinc chloride	62	+35	+40	3236	
n-Formyl-2-(Nitromethylene)-1,3-Perhydrothiazine	100	+45	+50	3236	
2-(2-Hydroxyethoxy)-1-(pyrrolidin-1-yl) Benzene-4-diazonium zinc chloride	100	+45	+50	3236	
3-(2-Hydroxyethoxy)-4-(pyrrolidin-1-yl) Benzene diazonium zinc chloride	100	+40	+45	3236	
☐ 7-Methoxy-5-methyl-benzothiophen-2-yl) boronic acid	88-100			3230	7
2-(N,N-Methylaminoethylcarbonyl)-4-(3,4-Dimethyl-phenylsulphonyl) Benzenediazonium hydrogen sulfate	96	+45	+50	3236	
4-Methylbenzenesulphonylhydrazide	100			3226	
3-Methyl-4-(Pyrrolidin-1-yl) Benzenediazonium tetrafluoroborate	95	+45	+50	3234	
4-Nitrosophenol	100	+35	+40	3236	
Phosphorothioic acid, O-[(cyanophenyl methylene) azanyl] O,O-diethyl ester	82-91 (Z isomer)			3227	6
Self-reactive liquid, sample				3223	4
Self-reactive liquid, sample, temperature controlled				3233	4
Self-reactive solid, sample				3224	4
Self-reactive solid, sample, temperature controlled				3234	4
Sodium 2-Diazo-1-naphthol-4-Sulphonate	100			3226	
Sodium 2-Diazo-1-naphthol-5-Sulphonate	100			3226	
Tetramine Palladium (II) Nitrate	100	+30	+35	3234	

Notes:

1. *For a complete description of the classification of Azodicarbonamide formulations see 2.4.2.3.3.2 of the UN Model Regulations.*
2. *"Explosive" subsidiary hazard label required and consequently forbidden for transport by air under any circumstances.*
3. *With a compatible diluent having a boiling point of not less than 150°C.*
4. *Approval of the appropriate national authority required, see 3.4.1.2.5.*
5. △ *This entry applies to mixtures of esters of 2-Diazo-1-Naphthol-4-Sulphonic acid and 2-Diazo-1-Naphthol-5-Sulphonic acid meeting the criteria of 2.4.2.3.3.2 d) of the UN Model Regulations.*
6. *This entry applies to the technical mixture in n-butanol within the specified concentration limits of the (Z) isomer.*
7. ☐ *The technical compound with the specified concentration limits may contain up to 12% water and up to 1% organic impurities.*

C.2 Organic Peroxides (Division 5.2)

This list is based on paragraph 2.5.3.2.4 of the of the UN Model Regulations, with irrelevant material removed.

Allocation of new organic peroxides or new formulations of currently assigned organic peroxides to a generic entry should be made by the competent authority of the country of manufacture and notification sent to the competent authority of the country of destination if so required by it.

Notes:

1. *The UN Model Regulations contains a complete description of the classification of Division 5.2, Organic Peroxides.*
2. *Peroxides to be transported must fulfil the classification and the control and emergency temperature (derived from the SADT) as listed.*
3. *Organic peroxides not listed in Table C.2 are subject to classification approval by the appropriate national authority of the State in which the dangerous goods were manufactured (See 3.5.2.3.1).*

TABLE C.2
List of Currently Assigned Organic Peroxides in Packages

Organic Peroxide	Concentration (%)	Diluent Type A (%)*	Diluent Type B (%)**	Inert solid (%)	Water (%)	Control Tempe-rature (°C)	Emer-gency Tempe-rature (°C)	UN Number (Generic Entry)	Notes
Acetyl acetone peroxide	≤ 42	≥ 48			≥ 8			3105	2
Acetyl acetone peroxide	≤ 35	≥ 57			≥ 8			3107	32
Acetyl acetone peroxide	≤ 32 as a paste							3106	20
Acetyl cyclohexanesulphonyl peroxide	≤ 82				≥ 12	-10	0	Forbidden	3
Acetyl cyclohexanesulphonyl peroxide	≤ 32		≥ 68			-10	0	3115	
tert-Amyl hydroperoxide	≤ 88	≥ 6			≥ 6			3107	
tert-Amyl peroxyacetate	≤ 62	≥ 38						3105	
tert-Amyl peroxybenzoate	≤ 100							3103	
tert-Amyl peroxy-2-ethylhexanoate	≤ 100					+20	+25	3115	
tert-Amyl peroxy-2-ethylhexyl carbonate	≤ 100							3105	
tert-Amylperoxy isopropyl carbonate	≤ 77	≥ 23						3103	
tert-Amyl peroxyneodecanoate	≤ 77		≥ 23			0	+10	3115	
tert-Amyl peroxyneodecanoate	≤ 47	≥ 53				0	+10	3119	
tert-Amyl peroxypivalate	≤ 77		≥ 23			+10	+15	3113	
tert-Amylperoxy-3,5,5-trimethylhexanoate	≤ 100							3105	
tert-Butyl cumyl peroxide	> 42-100							3109	
tert-Butyl cumyl peroxide	≤ 52			≥ 48				3108	
n-Butyl-4,4-di-(tert-butylperoxy) valerate	≤ 52			≥ 48				3108	
n-Butyl-4,4-di-(tert-butylperoxy) valerate	> 52-100							3103	
tert-Butyl hydroperoxide	> 79-90				≥ 10			3103	13
tert-Butyl hydroperoxide	≤ 79				> 14			3107	13, 23
tert-Butyl hydroperoxide	≤ 80	≥ 20						3105	4, 13
tert-Butyl hydroperoxide	≤ 72				≥ 28			3109	13
tert-Butyl hydroperoxide with Di-tert-Butyl peroxide	< 82 + > 9				≥ 7			3103	13
tert-Butyl monoperoxymaleate	> 52-100							Forbidden	3
tert-Butyl monoperoxymaleate	≤ 52			≥ 48				3108	
tert-Butyl monoperoxymaleate	≤ 52 as a paste							3108	
tert-Butyl monoperoxymaleate	≤ 52	≥ 48						3103	
tert-Butyl peroxyacetate	> 52-77	≥ 23						Forbidden	3
tert-Butyl peroxyacetate	> 32-52	≥ 48						3103	
tert-Butyl peroxyacetate	≤ 32		≥ 68					3109	
tert-Butyl peroxybenzoate	> 77-100							3103	
tert-Butyl peroxybenzoate	> 52-77	≥ 23						3105	
tert-Butyl peroxybenzoate	≤ 52			≥ 48				3106	
tert-Butyl peroxybutyl fumarate	≤ 52	≥ 48						3105	
tert-Butyl peroxycrotonate	≤ 77	≥ 23						3105	
tert-Butyl peroxydiethylacetate	≤ 100					+20	+25	3113	

TABLE C.2
List of Currently Assigned Organic Peroxides in Packages *(continued)*

Organic Peroxide	Concentration (%)	Diluent Type A (%)*	Diluent Type B (%)**	Inert solid (%)	Water (%)	Control Temperature (°C)	Emergency Temperature (°C)	UN Number (Generic Entry)	Notes
tert-Butyl peroxy-2-ethylhexanoate	> 52-100					+20	+25	3113	
tert-Butyl peroxy-2-ethylhexanoate	> 32-52		≥ 48			+30	+35	3117	
tert-Butyl peroxy-2-ethylhexanoate	≤ 52			≥ 48		+20	+25	3118	
tert-Butyl peroxy-2-ethylhexanoate	≤ 32		≥ 68			+40	+45	3119	
tert-Butyl peroxy-2-ethylhexanoate with 2,2-Di-(tert-butylperoxy) butane	≤ 31 + ≤ 36		≥ 33			+35	+40	3115	
tert-Butyl peroxy-2-ethylhexanoate with 2,2-Di-(tert-butylperoxy) butane	≤ 12 + ≤ 14	≥ 14		≥ 60				3106	
tert-Butyl peroxy-2-ethylhexylcarbonate	≤ 100							3105	
tert-Butyl peroxyisobutyrate	> 52-77		≥ 23			+15	+20	Forbidden	3
tert-Butyl peroxyisobutyrate	≤ 52		≥ 48			+15	+20	3115	
tert-Butylperoxy isopropylcarbonate	≤ 77	≥ 23						3103	
tert-Butylperoxy isopropylcarbonate	≤ 62		≥ 38					3105	
1-(2-tert-Butylperoxy isopropyl)-3-isopropenylbenzene	≤ 77	≥ 23						3105	
1-(2-tert-Butylperoxy isopropyl)-3-isopropenylbenzene	≤ 42			≥ 58				3108	
tert-Butyl peroxy-2-methylbenzoate	≤ 100							3103	
tert-Butyl peroxyneodecanoate	> 77-100					-5	+5	3115	
tert-Butyl peroxyneodecanoate	≤ 77		≥ 23			0	+10	3115	
tert-Butyl peroxyneodecanoate	≤ 52 as a stable dispersion in water					0	+10	3119	
tert-Butyl peroxyneodecanoate	≤ 42 as a stable dispersion in water (frozen)					0	+10	3118	
tert-Butyl peroxyneodecanoate	≤ 32	≥ 68				0	+10	3119	
tert-Butyl peroxyneoheptanoate	≤ 77	≥ 23				0	+10	3115	
tert-Butyl peroxyneoheptanoate	≤ 42 as a stable dispersion in water					0	+10	3117	
tert-Butyl peroxypivalate	> 67-77	≥ 23				0	+10	3113	
tert-Butyl peroxypivalate	> 27-67		≥ 33			0	+10	3115	
tert-Butyl peroxypivalate	≤ 27		≥ 73			+30	+35	3119	
tert-Butylperoxy stearylcarbonate	≤ 100							3106	
tert-Butyl peroxy-3,5,5-trimethylhexanoate	> 37-100							3105	
tert-Butyl peroxy-3,5,5-trimethylhexanoate	≤ 42			≥ 58				3106	
tert-Butyl peroxy-3,5,5-trimethylhexanoate	≤ 37		≥ 63					3109	
3-Chloroperoxybenzoic acid	> 57-86			≥ 14				Forbidden	3
3-Chloroperoxybenzoic acid	≤ 77			≥ 6	≥ 17			3106	
3-Chloroperoxybenzoic acid	≤ 57			≥ 3	≥ 40			3106	
Cumyl hydroperoxide	> 90-98	≤ 10						3107	13
Cumyl hydroperoxide	≤ 90	≥ 10						3109	13, 18
Cumyl peroxyneodecanoate	≤ 52 as a stable dispersion in water					-10	0	3119	
Cumyl peroxyneodecanoate	≤ 77		≥ 23			-10	0	3115	
Cumyl peroxyneodecanoate	≤ 87	≥ 13				-10	0	3115	
Cumyl peroxyneoheptanoate	≤ 77	≥ 23				-10	0	3115	
Cumyl peroxypivalate	≤ 77		≥ 23			-5	+5	3115	
Cyclohexanone peroxide(s)	≤ 91				≥ 9			3104	13
Cyclohexanone peroxide(s)	≤ 72 as a paste							3106	5, 20
Cyclohexanone peroxide(s)	≤ 72	≥ 28						3105	5
Cyclohexanone peroxide(s)	≤ 32			≥ 68				Exempt	29

TABLE C.2
List of Currently Assigned Organic Peroxides in Packages *(continued)*

Organic Peroxide	Concentration (%)	Diluent Type A (%)*	Diluent Type B (%)**	Inert solid (%)	Water (%)	Control Tempe-rature (°C)	Emer-gency Tempe-rature (°C)	UN Number (Generic Entry)	Notes
([3r-(3r,5as,6s,8as,9r,10r,12s,12ar**)]-Decahydro-10-methoxy-3,6,9- trimethyl-3,12-epoxy-12h-pyrano[4,3- j]-1,2-benzodioxepin)	≤ 100							3106	
Diacetone alcohol peroxides	≤ 57		≥ 26		≥ 8	+40	+45	3115	6
Diacetyl peroxide	≤ 27		≥ 73			+20	+25	3115	7, 13
Di-tert-Amyl peroxide	≤ 100							3107	
2,2-Di-(tert-amylperoxy) butane	≤ 57	≥ 43						3105	
1,1-Di-(tert-amylperoxy) cyclohexane	≤ 82	≥ 18						3103	
Dibenzoyl peroxide	> 36-42	≥ 18			≤ 40			3107	
Dibenzoyl peroxide	> 52-100			≤ 48				Forbidden	3
Dibenzoyl peroxide	> 77-94				≥ 6			Forbidden	3
Dibenzoyl peroxide	≤ 77				≥ 23			3104	
Dibenzoyl peroxide	≤ 62			≥ 28	≥ 10			3106	
Dibenzoyl peroxide	≤ 56.5 as a paste				≥ 15			3108	
Dibenzoyl peroxide	> 52-62 as a paste							3106	20
Dibenzoyl peroxide	≤ 52 as a paste							3108	20
Dibenzoyl peroxide	> 35-52			≥ 48				3106	
Dibenzoyl peroxide	≤ 42 as a stable dispersion in water							3109	
Dibenzoyl peroxide	≤ 35			≥ 65				Exempt	29
Di-(4-tert-butylcyclohexyl) peroxydicarbonate	≤ 100					+30	+35	3114	
Di-(4-tert-butylcyclohexyl) peroxydicarbonate	≤ 42 as a paste					+35	+40	3118	
Di-(4-tert-butylcyclohexyl) peroxydicarbonate	≤ 42 as a stable dispersion in water					+30	+35	3119	
Di-tert-butyl peroxide	> 52-100							3107	
Di-tert-butyl peroxide	≤ 52		≥ 48					3109	25
Di-tert-butyl peroxyazelate	≤ 52	≥ 48						3105	
2,2-Di-(tert-butylperoxy) butane	≤ 52	≥ 48						3103	
1,6-Di-(tert-butylperoxycarbonyloxy) hexane	≤ 72	28≥						3103	
1,1-Di-(tert-butylperoxy) cyclohexane	> 80-100							Forbidden	3
1,1-Di-(tert-butylperoxy) cyclohexane	≤ 72		≥ 28					3103	30
1,1-Di-(tert-butylperoxy) cyclohexane	> 52-80	≥ 20						3103	
1,1-Di-(tert-butylperoxy) cyclohexane	> 42-52	≥ 48						3105	
1,1-Di-(tert-butylperoxy) cyclohexane	≤ 42	≥ 13		≥ 45				3106	
1,1-Di-(tert-butylperoxy) cyclohexane	≤ 27	≥ 25						3107	21
1,1-Di-(tert-butylperoxy) cyclohexane	≤ 42	≥ 58						3109	
1,1-Di-(tert-butylperoxy) cyclohexane	≤ 13	≥ 13	≥ 74					3109	
1,1-Di-(tert-butylperoxy) cyclohexane with tert-butyl peroxy-2-ethylhexanoate	≤ 43 + ≤ 16	≥ 41						3105	
1,1-Di-(tert-butylperoxy)-3,3,5-trimethylcyclohexane	≤ 90		≥ 10					3103	30
Di-n-butyl peroxydicarbonate	> 27-52		≥ 48			-15	-5	3115	
Di-n-butyl peroxydicarbonate	≤ 27		≥ 73			-10	0	3117	
Di-n-butyl peroxydicarbonate	≤ 42 as a stable dispersion in water (frozen)					-15	-5	3118	
Di-sec-butyl peroxydicarbonate	> 52-100					-20	-10	3113	
Di-sec-butyl peroxydicarbonate	≤ 52		≥ 48			-15	-5	3115	
Di-(tert-butylperoxyisopropyl) benzene(s)	> 42-100			≤ 57				3106	
Di-(tert-butylperoxyisopropyl) benzene(s)	≤ 42			≥ 58				Exempt	29
Di-(tert-butylperoxy) phthalate	> 42-52	≥ 48						3105	

TABLE C.2
List of Currently Assigned Organic Peroxides in Packages *(continued)*

Organic Peroxide	Concentration (%)	Diluent Type A (%)*	Diluent Type B (%)**	Inert solid (%)	Water (%)	Control Temperature (°C)	Emergency Temperature (°C)	UN Number (Generic Entry)	Notes
Di-(tert-butylperoxy) phthalate	≤ 52 as a paste							3106	20
Di-(tert-butylperoxy) phthalate	≤ 42	≥ 58						3107	
2,2-Di-(tert-butylperoxy) propane	≤ 52	≥ 48						3105	
2,2-Di-(tert-butylperoxy) propane	≤ 42	≥ 13		≥ 45				3106	
1,1-Di-(tert-butylperoxy)-3,3,5-trimethylcyclohexane	> 90-100							Forbidden	3
1,1-Di-(tert-butylperoxy)-3,3,5-trimethylcyclohexane	> 57-90	≥ 10						3103	
1,1-Di-(tert-butylperoxy)-3,3,5-trimethylcyclohexane	≤ 90		≥ 10					3103	30
1,1-Di-(tert-butylperoxy)-3,3,5-trimethylcyclohexane	≤ 77		≥ 23					3103	
1,1-Di-(tert-butylperoxy)-3,3,5-trimethylcyclohexane	≤ 57			≥ 43				3110	
1,1-Di-(tert-butylperoxy)-3,3,5-trimethylcyclohexane	≤ 57	≥ 43						3107	
1,1-Di-(tert-butylperoxy)-3,5,5-trimethylcyclohexane	≤ 32	≥ 26	≥ 42					3107	
Dicetyl peroxydicarbonate	≤ 100					+30	+35	3120	
Dicetyl peroxydicarbonate	≤ 42 as a stable dispersion in water					+30	+35	3119	
Di-4-chlorobenzoyl peroxide	≤ 77				≥ 23			Forbidden	3
Di-4-chlorobenzoyl peroxide	≤ 52 as a paste							3106	20
Di-4-chlorobenzoyl peroxide	≤ 32			≥ 68				Exempt	29
Dicumyl peroxide	> 52-100							3110	12
Dicumyl peroxide	≤ 52			≥ 48				Exempt	29
Dicyclohexyl peroxydicarbonate	> 91-100					+10	+15	Forbidden	3
Dicyclohexyl peroxydicarbonate	≤ 91				≥ 9	+10	+15	3114	
Dicyclohexyl peroxydicarbonate	≤ 42 as a stable dispersion in water					+15	+20	3119	
Didecanoyl peroxide	≤ 100					+30	+35	3114	
2,2-Di(4,4-di-(tert-butylperoxy) cyclohexyl) propane	≤ 42			≥ 58				3106	
2,2-Di(4,4-di-(tert-butylperoxy) cyclohexyl) propane	≤ 22		≥ 78					3107	
Di-2,4-dichlorobenzoyl peroxide	≤ 77				≥ 23			Forbidden	3
Di-2,4-dichlorobenzoyl peroxide	≤ 52 as a paste					+20	+25	3118	
Di-2,4-dichlorobenzoyl peroxide	≤ 52 as a paste with silicon oil							3106	
Di-(2-ethoxyethyl) peroxydicarbonate	≤ 52		≥ 48			-10	0	3115	
Di-(2-ethylhexyl) peroxydicarbonate	> 77-100					-20	-10	3113	
Di-(2-ethylhexyl) peroxydicarbonate	≤ 77		≥ 23			-15	-5	3115	
Di-(2-ethylhexyl) peroxydicarbonate	≤ 62 as a stable dispersion in water					-15	-5	3119	
Di-(2-ethylhexyl) peroxydicarbonate	≤ 52 as a stable dispersion in water (frozen)					-15	-5	3120	
2,2-Dihydroperoxypropane	≤ 27			≥ 73				Forbidden	3
Di-(1-hydroxycyclohexyl) peroxide	≤ 100							3106	
Diisobutyryl peroxide	> 32-52		≥ 48			-20	-10	Forbidden	3
Diisobutyryl peroxide	≤ 42 as a stable dispersion in water					-20	-10	3119	
Diisobutyryl peroxide	≤ 32		≥ 68			-20	-10	3115	
Diisopropylbenzene dihydroperoxide	≤ 82	≥ 5			≥ 5			3106	24
Diisopropyl peroxydicarbonate	> 52-100					-15	-5	Forbidden	3
Diisopropyl peroxydicarbonate	≤ 52		≥ 48			-20	-10	3115	
Diisopropyl peroxydicarbonate	≤ 32	≥ 68				-15	-5	3115	
Dilauroyl peroxide	≤ 100							3106	
Dilauroyl peroxide	≤ 42 as a stable dispersion in water							3109	

TABLE C.2
List of Currently Assigned Organic Peroxides in Packages *(continued)*

Organic Peroxide	Concentration (%)	Diluent Type A (%)*	Diluent Type B (%)**	Inert solid (%)	Water (%)	Control Temperature (°C)	Emergency Temperature (°C)	UN Number (Generic Entry)	Notes
Di-(3-methoxybutyl) peroxydicarbonate	≤ 52		≥ 48			-5	+5	3115	
Di-(2-methylbenzoyl) peroxide	≤ 87				≥ 13	+30	+35	Forbidden	3
Di-(4-methylbenzoyl) peroxide	≤ 52 as a paste with silicon oil							3106	
Di-(3-methylbenzoyl) peroxide + Benzoyl (3-methylbenzoyl) peroxide + dibenzoyl peroxide	≤ 20 + ≤ 18 + ≤ 4		≥ 58			+35	+40	3115	
2,5-Dimethyl-2,5-di-(benzoylperoxy) hexane	> 82-100							Forbidden	3
2,5-Dimethyl-2,5-di-(benzoylperoxy) hexane	≤ 82			≥ 18				3106	
2,5-Dimethyl-2,5-di-(benzoylperoxy) hexane	≤ 82				≥ 18			3104	
2,5-Dimethyl-2,5-di-(tert-butylperoxy) hexane	> 90-100							3103	
2,5-Dimethyl-2,5-di-(tert-butylperoxy) hexane	> 52-90	≥ 10						3105	
2,5-Dimethyl-2,5-di-(tert-butylperoxy) hexane	≤ 52	≥ 48						3109	
2,5-Dimethyl-2,5-di-(tert-butylperoxy) hexane	≤ 77			≥ 23				3108	
2,5-Dimethyl-2,5-di-(tert-butylperoxy) hexane	≤ 47 as a paste							3108	
2,5-Dimethyl-2,5-di-(tert-butylperoxy) hexyne-3	> 86-100							Forbidden	3
2,5-Dimethyl-2,5-di-(tert-butylperoxy) hexyne-3	> 52-86	≥ 14						3103	26
2,5-Dimethyl-2,5-di-(tert-butylperoxy) hexyne-3	≤ 52			≥ 48				3106	
2,5-Dimethyl-2,5-di-(2-ethylhexanoylperoxy) hexane	≤ 100					+20	+25	3113	
2,5-Dimethyl-2,5-dihydroperoxyhexane	≤ 82				≥ 18			3104	
2,5-Dimethyl-2,5-di-(3,5,5-trimethylhexanoylperoxy) hexane	≤ 77	≥ 23						3105	
1,1-Dimethyl-3-hydroxybutyl peroxyneoheptanoate	≤ 52	≥ 48				0	+10	3117	
Dimyristyl peroxydicarbonate	≤ 100					+20	+25	3116	
Dimyristyl peroxydicarbonate	< 42 as a stable dispersion in water					+20	+25	3119	
Di-(2-neodecanoylperoxyisopropyl) benzene	≤ 52	≥ 48				-10	0	3115	
Di-n-nonanoyl peroxide	≤ 100					0	+10	3116	
Di-n-octanoyl peroxide	≤ 100					+10	+15	3114	
Di-(2-phenoxyethyl) peroxydicarbonate	> 85-100							Forbidden	3
Di-(2-phenoxyethyl) peroxydicarbonate	≤ 85				≥ 15			3106	
Dipropionyl peroxide	≤ 27		≥ 73			+15	+20	3117	
Di-n-propyl peroxydicarbonate	≤ 100					-25	-15	3113	
Di-n-propyl peroxydicarbonate	≤ 77	≥ 23				-20	-10	3113	
Disuccinic acid peroxide	> 72-100							Forbidden	3, 17
Disuccinic acid peroxide	≤ 72				≥ 28	+10	+15	3116	
Di-(3,5,5-trimethylhexanoyl) peroxide	> 52-82	≥ 18				0	+10	3115	
Di-(3,5,5-trimethylhexanoyl) peroxide	≤ 52 as a stable dispersion in water					+10	+15	3119	
Di-(3,5,5-trimethylhexanoyl) peroxide	> 38-52	≥ 48				+10	+15	3119	
Di-(3,5,5-trimethylhexanoyl) peroxide	≤ 38	≥ 62				+20	+25	3119	
Ethyl 3,3-di-(tert-amylperoxy) butyrate	≤ 67	≥ 33						3105	
Ethyl 3,3-di-(tert-butylperoxy) butyrate	> 77-100							3103	
Ethyl 3,3-di-(tert-butylperoxy) butyrate	≤ 77	≥ 23						3105	
Ethyl 3,3-di-(tert-butylperoxy) butyrate	≤ 52			≥ 48				3106	
1-(2-Ethylhexanoylperoxy)-1,3-dimethylbutyl peroxypivalate	≤ 52	≥ 45	≥ 10			-20	-10	3115	
tert-Hexyl Peroxyneodecanoate	≤ 71	≥ 29				0	+10	3115	
tert-Hexyl peroxypivalate	≤ 72		≥ 28			+10	+15	3115	
☐ tert-Hexyl peroxypivalate	≤ 52 as a stable dispersion in water					+15	+20	3117	

TABLE C.2
List of Currently Assigned Organic Peroxides in Packages *(continued)*

Organic Peroxide	Concentration (%)	Diluent Type A (%)*	Diluent Type B (%)**	Inert solid (%)	Water (%)	Control Tempe-rature (°C)	Emer-gency Tempe-rature (°C)	UN Number (Generic Entry)	Notes
3-Hydroxy-1,1-dimethylbutyl peroxyneodecanoate	≤ 77	≥ 23				-5	+5	3115	
3-Hydroxy-1,1-dimethylbutyl peroxyneodecanoate	≤ 52 as a stable dispersion in water					-5	+5	3119	
3-Hydroxy-1,1-dimethylbutyl peroxyneodecanoate	≤ 52	≥ 48				-5	+5	3117	
Isopropyl sec-butyl peroxydicarbonate + di-sec-butyl peroxydicarbonate + di-isopropyl peroxydicarbonate	≤ 32 + ≤ 15 - 18 + ≤ 12 - 15	≥ 38				-20	-10	3115	
Isopropyl sec-butyl peroxydicarbonate + di-sec-butyl peroxydicarbonate + di-isopropyl peroxydicarbonate	≤ 52 + ≤ 28 + ≤ 22					-20	-10	Forbidden	3
Isopropylcumyl hydroperoxide	≤ 72	≥ 28						3109	13
p-Menthyl hydroperoxide	> 72-100							3105	13
p-Menthyl hydroperoxide	≤ 72	≥ 28						3109	27
Methylcyclohexanone peroxide(s)	≤ 67		≥ 33			+35	+40	3115	
Methyl ethyl ketone peroxide(s)	(see Note 8)	≥ 48						Forbidden	3, 8, 13
Methyl ethyl ketone peroxide(s)	(see Note 9)	≥ 55						3105	9
Methyl ethyl ketone peroxide(s)	(see Note 10)	≥ 60						3107	10
Methyl isobutyl ketone peroxide(s)	≤ 62	≥ 19						3105	22
Methyl isopropyl ketone peroxide(s)	See Note 31	≥ 70						3109	31
Organic peroxide, liquid, sample								3103	11
Organic peroxide, liquid, sample, temperature controlled								3113	11
Organic peroxide, solid, sample								3104	11
Organic peroxide, solid, sample, temperature controlled								3114	11
3,3,5,7,7-pentamethyl-1,2,4-trioxepane	≤ 100							3107	
Peroxyacetic acid, type D, stabilized	≤ 43							3105	13, 14, 19
Peroxyacetic acid, type E, stabilized	≤ 43							3107	13, 15, 19
Peroxyacetic acid, type F, stabilized	≤ 43							3109	13, 16, 19
Peroxylauric acid	≤ 100					+35	+40	3118	
1-Phenylethyl hydroperoxide	≤ 38		≥ 62					3109	
Pinanyl hydroperoxide	> 56-100							3105	13
Pinanyl hydroperoxide	≤ 56	≥ 44						3109	
Polyether poly-tert-butylperoxycarbonate	≤ 52		≥ 48					3107	
1,1,3,3-Tetramethylbutyl hydroperoxide	≤ 100							3105	
1,1,3,3-Tetramethylbutylperoxy-2 ethyl-hexanoate	≤ 100					+15	+20	3115	
1,1,3,3- Tetramethylbutyl peroxyneodecanoate	≤ 72		≥ 28			-5	+5	3115	
1,1,3,3- Tetramethylbutyl peroxyneodecanoate	≤ 52 as a stable dispersion in water					-5	+5	3119	
1,1,3,3-Tetramethylbutyl peroxypivalate	≤ 77	≥ 23				0	+10	3115	
△ 3,6,9-Triethyl-3,6,9-trimethyl-1,4,7 triperoxonane	≤ 17	≥ 18		≥ 65				3110	
□ 3,6,9-Triethyl-3,6,9-trimethyl-1,4,7 triperoxonane	≤ 27	≥ 83						3109	
3,6,9-Triethyl-3,6,9 trimethyl-1,4,7 triperoxonane	≤ 42	≥ 58						3105	28

* "DILUENTS TYPE A" are organic liquids which are compatible with the organic peroxide and which have a boiling point of not less than 150°C. Type A diluents may be used for desensitising all organic peroxides.

** "DILUENTS TYPE B" are organic liquids which are compatible with the organic peroxide and which have a boiling point of less than 150°C but not less than 60°C and a flashpoint of not less than 5°C. Type B diluents may only be used for desensitisation of organic peroxides for which the temperature control is required. The boiling point of the liquid should be at least 50°C higher than the control temperature of the organic peroxide.

> = greater than.

≥ = equal to or greater than.

< = less than.

≤ = equal to or less than.

Notes:

1. Diluent type B may always be replaced by diluent type A. Boiling point diluent type B should be at least 60°C higher than the SADT of the organic peroxide.
2. Available oxygen ≤ 4.7%.
3. "Explosive" subsidiary hazard label required (see Figure 7.3.B) and consequently forbidden for transport by air under any circumstances.
4. Diluent may be replaced by Di-tert-butyl peroxide.
5. Available oxygen ≤ 9%.
6. With ≤ 9% hydrogen peroxide; available oxygen ≤ 10%.
7. Only non-metallic packagings allowed.
8. Available oxygen > 10% and ≤ 10.7%, with or without water.
9. Available oxygen ≤ 10%, with or without water.
10. Available oxygen ≤ 8.2%, with or without water.
11. See 3.5.2.6.
12. Not used.
13. "Corrosive" subsidiary hazard label required (see Figure 7.3.V).
14. Peroxyacetic acid formulations which fulfil the criteria of organic peroxide type D.
15. Peroxyacetic acid formulations which fulfil the criteria of organic peroxide type E.
16. Peroxyacetic acid formulations which fulfil the criteria of organic peroxide type F.
17. Addition of water to this organic peroxide will decrease its thermal stability.
18. No "Corrosive" subsidiary hazard label required for concentrations below 80%.
19. Mixtures with hydrogen peroxide, water and acid(s).
20. With diluent type A, with or without water.
21. With ≥ 25% diluent type A by mass and in addition ethylbenzene.
22. With ≥ 19%, diluent type A by mass and in addition methyl isobutyl ketone.
23. With < 6% di-tert-butyl peroxide.
24. With ≤ 8% 1-isopropylhydroperoxy-4-isopropylhydroxy benzene.
25. Diluent Type B with boiling point > 110°C.
26. With < 0.5% hydroperoxides content.
27. For concentrations more than 56%, "Corrosive" subsidiary hazard label required (see Figure 7.3.V).
28. Available active oxygen ≤ 7.6% in diluent type A having a 95% boil-off point in the range of 220–260°C.
29. Not subject to the requirements of these Regulations for Division 5.2.
30. Diluent type B with boiling point > 130°C.
31. Active oxygen ≤ 6.7%.
32. ☐ Active oxygen ≤ 4.15%.